# Chipathon 2025
# Tech-8

Mosbius-Style Chips as an Alternative to Physical Lab Kits

# Tech-8 Team

## Team members:

Darren Johan <@monetaryape450>
Abdul Aziz
Muhammad Falih Rosyidi
Habibie Muhammad Ghifari

## Team background

- Academic Experience

  Recently completed the second year of study in Electrical Engineering at Institut Teknologi Bandung (ITB)

- Work Experience:

  No formal work experience yet; currently a full-time student

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# Tech8 Team

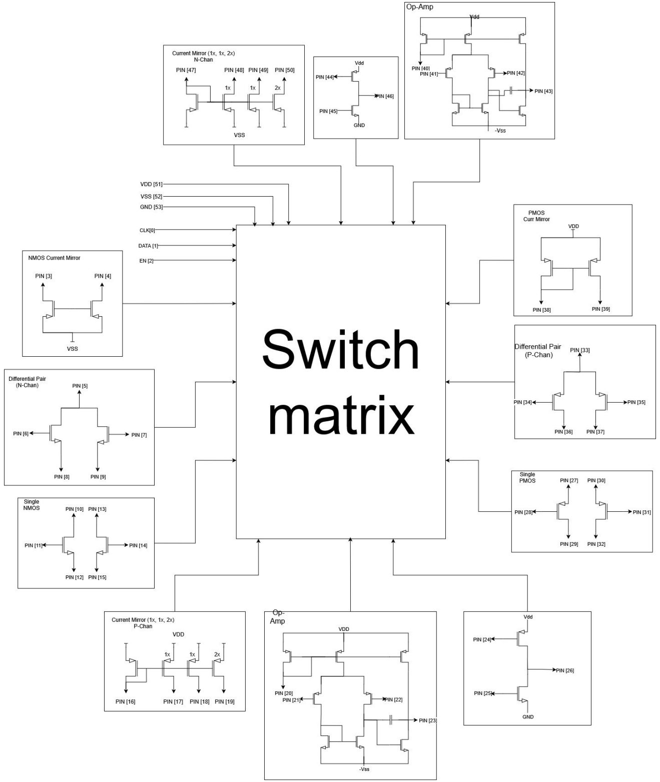
## Project information

Goal :

- Creating a (mostly) all-in-one chip to emulate fundamental properties of Transistors and ICs.

Applications :

- Emulating First Order HPF/LPF circuit with feedback.
- Characterization of Differential Pairs, Current mirrors, and other fundamental building blocks of ICS
- Characterization of single transistors and single-transistor amplifiers.

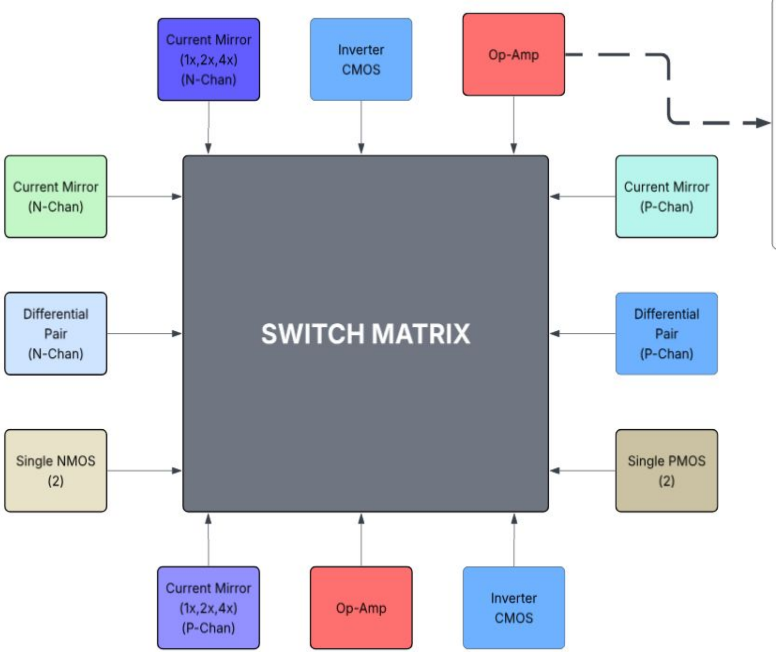


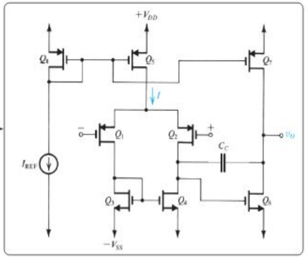

### Estimated transistor Count :

- Op-Amp : 8 * 2
- Current Mirror (P&N) : 2 * 2
- Differential Pair (P&N) : 2 * 2
- Single MOS (P&N) : 1 * 4
- CMOS Inverter : 2 * 2
- Current Mirror (Biasing,P&N) : 4 * 2

Total : ~ 40 Transistors

# Design Top level:



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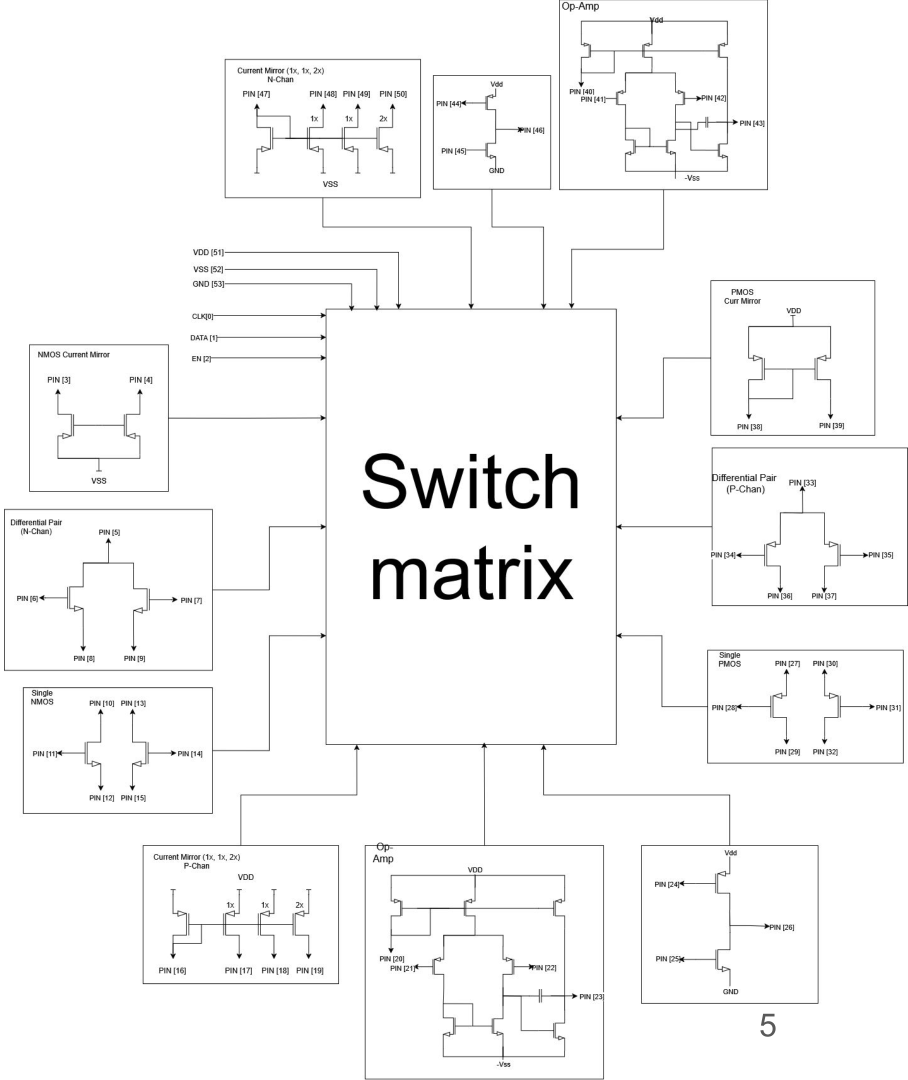
Link to Flowchart

# Design : [Link]()

| Pin | Type | Description |
|---|---|---|
| CLK | Digital Input | Switch Matrix Clock |
| DATA | Digital Input | Switch Matrix Data |
| EN | Digital Input | Switch Matrix Enable |
| GND | Power | Power Supply Ground |
| VSS | Power | Vss (-3.3V) |
| VDD | Power | Power Supply (3.3V) |
| 3 - 4 | Analog I/O | NMOS Current Mirror |
| 5 - 9 | Analog I/O | Differential Pair (N - Chan) |
| 10 - 15 | Analog I/O | Single NMOS |
| 16 - 19 | Analog I/O | Current Mirror (P-Chan) |
| 20 - 23 | Analog I/O | Op - Amp |
| 24 - 26 | Analog I/O | Inverter CMOS |
| 27 - 32 | Analog I/O | Single PMOS |
| 33 - 37 | Analog I/O | Differential Pair (P - Chan) |
| 38 - 39 | Analog I/O | PMOS Current Mirror |
| 40 - 43 | Analog I/O | Op Amp |
| 44 - 46 | Analog I/O | Inverter CMOS |
| 47 - 50 | Analog I/O | Current Mirror (N - Chan) |



# Task Division

- Switch Matrix and Top Integration
  - All Member
- Op Amp
  - Darren Johan
- Differential Pair & Current Mirror
  - Muhammad Falih Rosyidi
  - Habibie Muhammad Ghifari
- CMOS Pair & Current Mirror Biasing
  - Abdul Aziz

# Week by Week Schedule

| Project Phase | Week | Description | Work Deadlines |
|---|---|---|---|
| Team Formation and Project Planning | 27 | ● Team Formation<br>● Brainstorming<br>● Build Proposal | ● 05/07/2025 (Team formation)<br>● 13/07/2025 (Proposal Document) |
| Design and Simulation | 28 - 33 | ● Schematic Design<br>● Top-Level Blocks Simulation | ● 16/08/2025 (Go/No Go decision) |
| (Layout and Verification | 34 - 39 | ● Build Layout Blocks and Top Level<br>● DRC and LVS Check | ● 26/09/2025 (Final Submission) |
| Post Mortem | 40 | ● Documentation, Test plan Writing, Chip Usage Tutorials | |

# Tech8 Team

Questions, Suggestions, Doubts, :

- Best way to include the capacitor needed  for the op-amp, internal/external?
- Is the scope of the project okay, or do we need to reduce it ?

References :

[1] A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 7th ed. New York: Oxford University Press, 2015.

# TEMPLATE